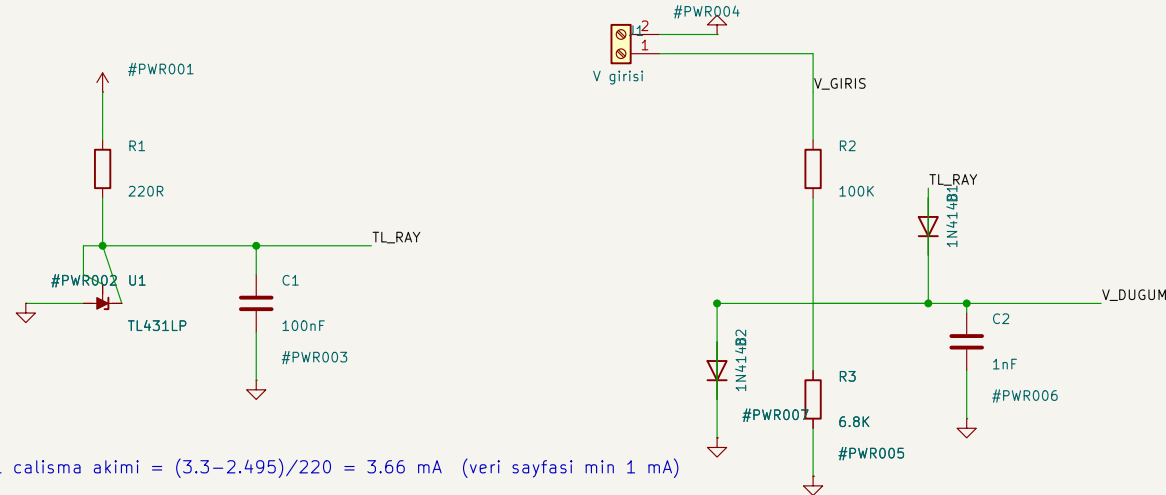


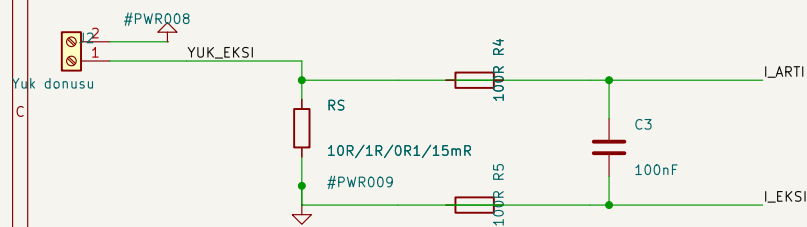
BLOK 1 – TL431 kelepce rayi 2.495 V (referans DEĞİL, KORUMA) BLOK 2 – Gerilim girişi, bolucu 15.71:1 → ADS #2 AINO



TL431 calisma akimi = $(3.3-2.495)/220 = 3.66 \text{ mA}$ (veri sayfasi min 1 mA)

Tam olcek 32.2 V @ PGA $+ - 2.048$ V, adim 982 uV. Dugum en fazla 2.048 V – kelepce 2.495 V'ta oldugu icin diyot TUM MENZILDE ters kutuplu, sizinti yok.

BLOK 3 – Sont + KELVIN algılama + diferansiyel RC → ADS #1 AIN0/AIN1



*** KELVIN – MONTAJ KURALI ***

R4/R5 algılama uçları sontun GOVDESİNE ayrı lehimlenecek, akim izine DEĞİL.

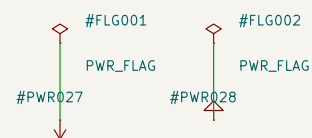
15 mohm kademesinde tek lehim noktasi (1 mohm) %6.7 hata demektir;

statik kısmi kalibrasyonla silinir ama 10 A'de yuke bagli %0.26 kalir.

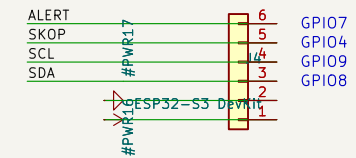
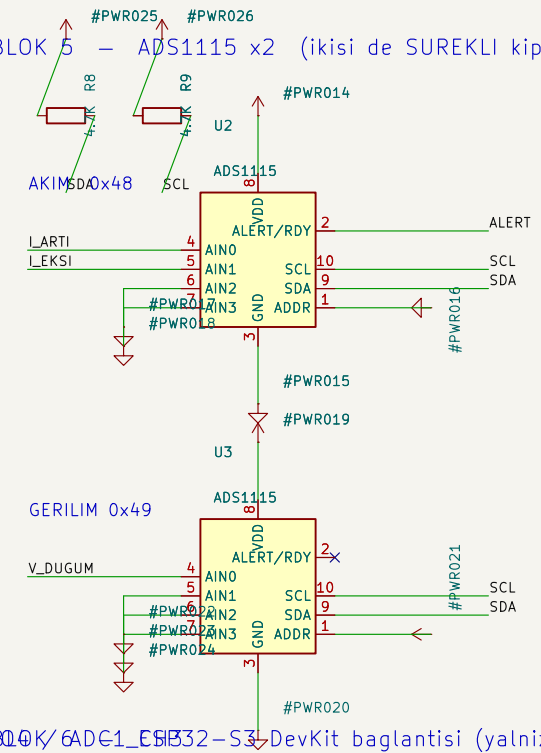
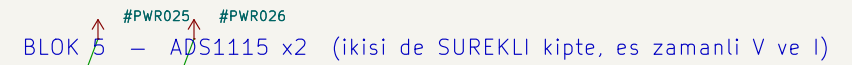
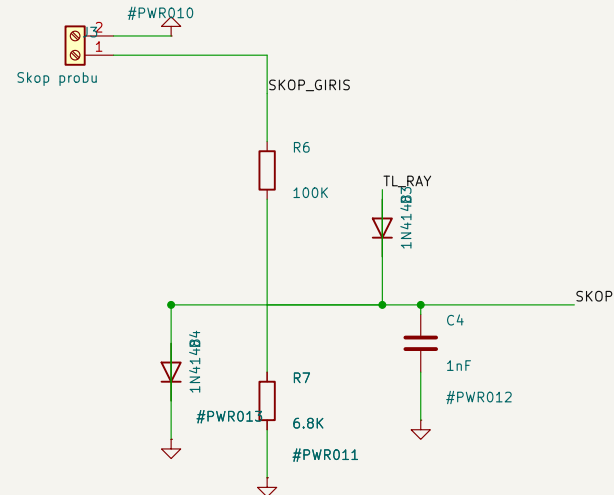
Asama 2 – tum degerler uretim/tasarim2.py (24/24) ve uretim/sim2_giris.py (11/11) ile dogrulandi.

Bolucu 15.71:1 SECIMDIR: 11:1 olsaydi TL431 kelepci 29.7 V'ta sizip menzili keserdi (A2, secenek C).

ADS1115 3.3 V'ta besleniyor – 5 V'ta VIH 3.5 V olur ve ESP32'nin 3.3 V'luk cikisi yetmezdi.



BLOK 4 – Osiloskop girişi (AYRI kanal) → ESP32 GND/6ADG1-ESP32-S3 DevKit bağlantısı (yalnız kullanılan pinler)



tasarim2.py + sim2_giris.py + netlist2_dogrulap.py ile dogrulandi Voltmetre 0-32.2 V / 982 uV · Ampermetre 0.78 uA – 11.5 A · Osiloskop 83 kSa/s 12 bit		
Elektronik Calisma Alani		
Sheet: / File: olcum-karti-a2.kicad_sch		
Title: Olcum Karti – Asama 2 (ESP32-S3 + ADS1115)		
Size: A3	Date:	Rev: 2
KiCad E.D.A. 10.0.6		Id: 1/1